Fe-C Problem 2 | Problems T4_PhaseDiagrams II.pdf #2 | Fe-C · eutectoid steel

Phases and Microconstituents of a Eutectoid Steel

Original statement · Problems T4_PhaseDiagrams II.pdf

2.- For a steel of C_0 = 0.76%C, at the following temperatures: T_1 = (1147+ΔT)ºC; T_2 = (1147-ΔT)ºC; T_3 = 900ºC; T_4 = (727+ΔT)ºC; T_5 = (727-ΔT)ºC. a) Indicate the phases present, their chemical composition and their mass fraction. b) Indicate the micro constituents, their chemical composition and their mass fraction at T_5.

Official Solution: a) T_1, T_2, T_3 and T_4: C_γ = 0.76% C, w_γ = 1; T_5: C_α = 0.022% C, C_{Fe_3C} = 6.7% C, w_α = 0.89, $w_{Fe_3C'}$ = 0.11; b) pearlite $w_{\alpha,E}$ = 0.89, $w_{Fe_3C,E}$ = 0.11

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium cooling on the Fe-Fe3C diagram: α 0.022, eutectoid 0.76 % C at 727 °C, Fe3C 6.67 % C [Session 9 Slide 4].
2. $C_0 = 0.76$ % C is the **eutectoid composition** [Session 9 Slide 12]: the alloy is austenite down to 727 °C (it touches the eutectoid point only there).

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C), one tie line per temperature; no change of basis. The eutectoid reaction $\gamma\,(0.76) \rightarrow \alpha\,(0.022) + \mathrm{Fe_3C}\,(6.67)$ turns all the austenite into pearlite at once [Session 9 Slide 10].

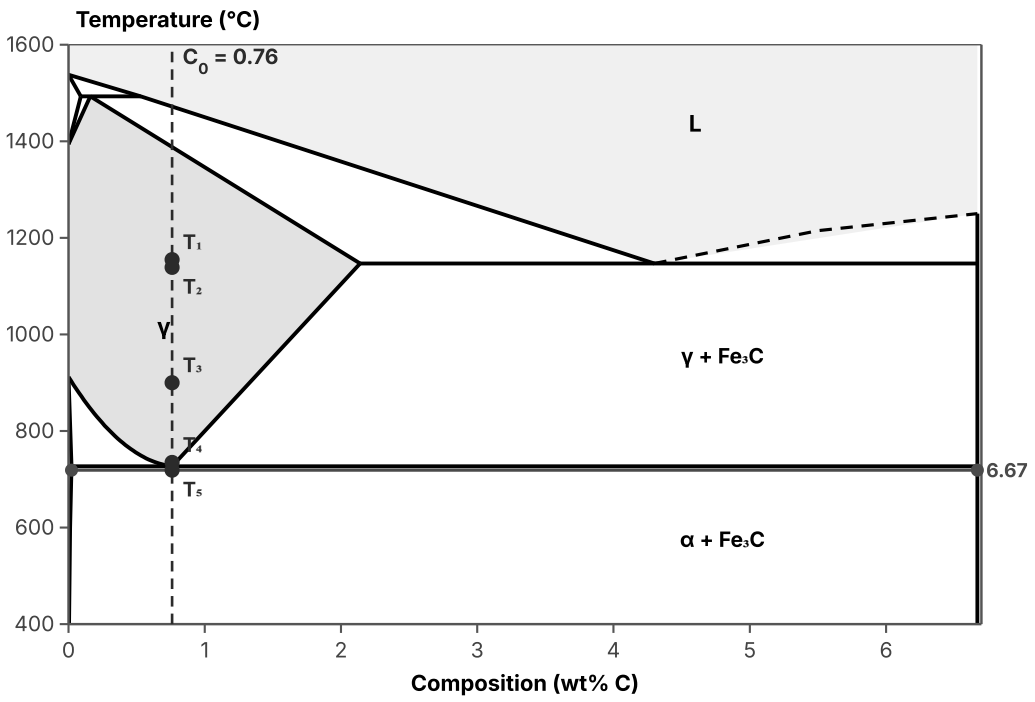


Figure 29. Fe-0.76 % C: γ down to 727 °C, then pearlite (α + Fe_3C).

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases T_1 to T_4: austenite, $C_\gamma = 0.76$ % C, $W_\gamma = 1$.

T_5 (length $6.67 - 0.022 = 6.648$):

$$W_\alpha = \frac{6.67 - 0.76}{6.648} = \frac{5.91}{6.648} = 0.889 \qquad W_{\mathrm{Fe_3C}} = \frac{0.76 - 0.022}{6.648} = \frac{0.738}{6.648} = 0.111$$

b) Microconstituents at T_5 One microconstituent, **pearlite**, 100 %, with overall composition 0.76 % C; inside it, eutectoid ferrite 0.889 and eutectoid cementite 0.111.

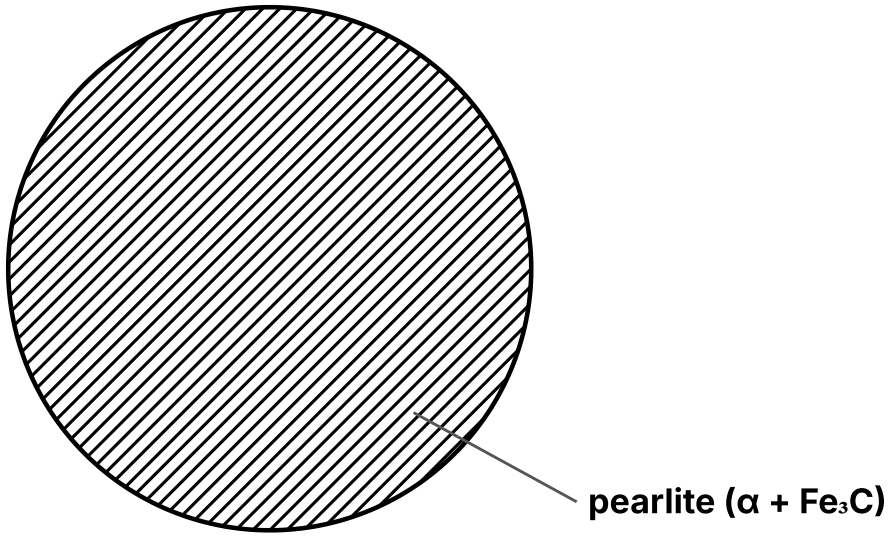


Figure 30. Fe-0.76 % C below 727 °C: 100 % pearlite.

PHASE 4 · CHECKS & INTERPRETATION

- Mass balance: $0.889 \times 0.022 + 0.111 \times 6.67 = 0.020 + 0.740 = 0.76$ % C .
- This is the make-up of every pearlite, the same as Worked example 15 of the theory (Session 9 Slide 13: 89 % and 11 %).
- **Limiting case** (fractions dimensionless, compositions in wt% C): at exactly 0.76 % C the proeutectoid fraction $(0.76 - C_0)/(0.76 - 0.022)$ is 0, the end case of the hypoeutectoid formula: the steel is all pearlite.

Check against the official solution

Agrees with the key (0.89 and 0.11). The key calls the cementite of part (a) "$w_{Fe_3C'}$" (with a prime, the symbol of proeutectoid cementite); in a eutectoid steel there is no proeutectoid cementite, all of it is in the pearlite, as part (b) of the key says.